Distributed Processing of GHCN-Daily Data with Apache Spark

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***Abstract*—Comprehending climate over time requires processing big datasets that exceed the capabilities of only one computer machine, necessitating distributed computing solutions for climate science applications. The main objective of this project is developing and deploying an automated distributed data processing pipeline to analyze climate anomaly patterns using the GHCN-Daily dataset. The pipeline integrates Amazon S3 for blob storage, Apache Spark for distributed processing, and MongoDB for analytical storage to examine three key research questions: how temperature anomalies have evolved across decades, what shifts have occurred in the ratio of heat to cold anomalies over time, and what statistical relationships exist between temperature and precipitation patterns. Using data from 70000 climate stations, we calculated how much temperatures differed from the 1980 to 2010 period. The key findings suggest a change from cold to heat climate patterns, with the heat to cold anomaly ratio increasing from 0.86 in the 1980s to 3.72 in the 2020s, indicating that extreme heat events now occur more frequently than extreme cold events. Statistical analysis reveals a moderate negative correlation with r=-0.42 and $p < 0.001$ between temperature and precipitation anomalies, suggesting changes in more than one climate variable.**

***Index Terms*—Big Data, AWS S3, Apache Spark, MongoDB, Climate**

I. Introduction

In today's world, climate change is one of the most serious challenges that humans are facing. Studies involving temperature trends and rainfall patterns require analysis of large datasets covering many decades and thousands of weather stations around the world. Traditional analysis methods have difficulty handling this amount of datasets in an efficient way, making necessary the use of distributed computing technologies.

This project presents a complete data pipeline for analyzing climate anomalies using the Global Historical Climatology Network Daily (GHCN-Daily) dataset. The records covering from the early 20th century to present day. To process over 70000 meteorological stations, we implement a scalable architecture combining cloud blob storage Amazon S3, distributed processing Apache Spark, and NoSQL databases MongoDB.

A. Research Question

The objective of this project is to identify temporal patterns in temperature anomalies by addressing three specific research questions:

RQ1: How has the frequency and intensity of temperature anomalies changed from 1980 to 2020?

RQ2: Is there a measurable shift in the ratio of heat anomalies to cold anomalies over time?

RQ3: What statistical relationship exists between temperature deviations and precipitation patterns in the same regions?

To answer these questions, we calculate temperature anomalies from 1980 to 2020. We implement the complete data pipeline using Python for data ingestion and orchestration, PySpark for distributed transformations and aggregations, and automated shell scripts to ensure reproducibility.

B. Importance of the Project

This project is important because it demonstrates a practical application of distributed computing to real-world climate problems and processing large semi-structured data formats using big data tools. Also, the automated pipeline architecture is generalizable to other large-scale temporal datasets requiring similar processing patterns.

C. Paper Organization

This paper is organized as follows: Section II reviews related work in climate data analysis and big data processing. Section III presents the methodology through architecture design and tools. Section IV presents the data processing pipeline including data acquisition, Spark processing, MongoDB storage, analytics and visualizations, and Dagster orchestration. Section V discusses findings, limitations, and future work.

II. Related Work

A. Climate Data Analysis Methodologies

The analysis of climate trends over time has been extensively studied in the scientific literature. Hansen et al. [1] established foundational methodologies for calculating temperature anomalies using 30-year baseline periods, demonstrating that this approach effectively captures climate variability while maintaining statistical significance. The World Meteorological Organization [3] formally adopted the 1981-2010 period as the standard climatological normal.

Several studies have examined global temperature trends using the GHCN-Daily dataset. Jones and Moberg [6] analyzed station-level temperature records to reconstruct hemispheric temperature variations, though their work predates modern

distributed computing capabilities. More recently, Menne et al. [2] described the GHCN-Daily dataset structure and quality control procedures, providing essential context for researchers utilizing this data source.

B. Big Data Technologies in Climate Science

The application of distributed computing frameworks to climate data analysis has gained prominence as dataset sizes exceed single machine processing capabilities. Dean and Ghemawat [4] introduced the MapReduce programming model, which forms the conceptual foundation for Apache Spark's distributed processing architecture. Zaharia et al. [5] presented Spark's resilient distributed datasets (RDDs), demonstrating significant performance improvements over traditional MapReduce for iterative algorithms common in data analysis tasks.

Several researchers have applied Spark to meteorological data processing, achieving substantial performance gains through parallelization. However, previous work focused on structured gridded data rather than the semi-structured station-based format of GHCN-Daily.

C. Research Gap

While existing literature demonstrates both climate analysis methodologies and distributed computing applications, few studies present complete pipelines processing raw GHCN-Daily station data at scale. Previous works either (a) analyze pre-aggregated climate indices rather than station-level data, (b) process data on single machines without distributed computing, or (c) focus on technical performance rather than climate science questions.

This work addresses this gap by implementing a complete automated pipeline from raw data ingestion through distributed processing to analytical storage, while simultaneously addressing substantive climate research questions regarding anomaly patterns and precipitation relationships.

III. Methodology

The methodology is an essential part of any project, it ensures the effectiveness of the proposed project and contributes to the successful achievement of the project objectives.

A. Project Overview

This section provides a comprehensive flowchart pipeline that implements many stages following Extract-Transform-Load (ETL) principles, integrating cloud storage, distributed processing, and analytical databases. Figure 1 illustrates the complete data flow from raw data acquisition through final visualization

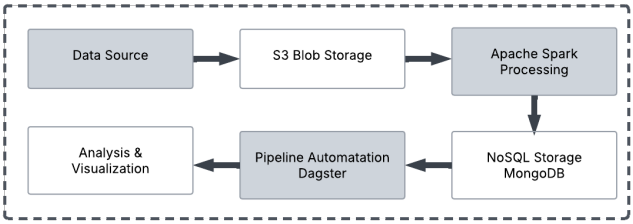

Fig. 1: Flowchart of the project process.

B. Technologies

This project was done on a personal laptop using the big data tools:

1) Amazon S3 Blob Storage: The storage solution provides scalable cloud with native integration into Apache Spark via s3a:// interface. It is designed to support operations on large files.

2) Apache Spark 3.5.1: The Apache Spark handles semi-structured data and hold in-memory processing to optimize iterative algorithms. The framework enables parallel processing across multiple files and provides a SQL-like API that simplifies data transformation and analysis.

3) MongoDB 7.0: MongoDB is a flexible NoSQL database that supports schema storage for analytical results. Its document-oriented data model is suited for storing aggregated data, and it provides a native Spark connector that enables direct writes from distributed processing jobs.

4) Python 3.x: Python is used to interact with Amazon S3 through the 'boto3' library and to implement distributed processing using the PySpark API. It also offers analytical and visualization libraries, such as pandas and matplotlib, which support data analysis.

C. Dagster

The complete data processing workflow is managed using Dagster, which treats data assets as core parts of the system.

IV. Data Processing Pipeline

A data processing pipeline is an important step for automation, transformation, and transfer of data from diverse sources to a single destination. It ensures faster and reduced manual errors, which is essential for data scalability. In this section, the data processing pipeline provides a comprehensive list of stages used during the process that can be followed to replicate the results produced in this proposed project.

A. Stage 1: Large-Scale Dataset

The latest available GHCN-Daily weather station dataset was obtained from NOAA's National Centers for Environmental Information link: ftp://ftp.ncdc.noaa.gov/pub/data/ghcn/daily. The ghcnd-stations.txt and ghcnd-inventory.txt files are also required, as they contain relevant information on station location and activity.

B. Stage 2: Data Organization and Blob Storage

Before starting data processing, we created a clear directory organization for scripts, data, and logs. Then, we configured a Python virtual environment with all permission dependencies, including boto3 for S3 access, PySpark for distributed processing, and pymongo for database connection. AWS credentials were set for programmatic access to the S3 bucket. MongoDB was installed and started as a local service.

Finally, we developed the automation scripts that make up the data pipeline. This careful preparation ensured that all the infrastructure was ready before starting large data processing at scale.

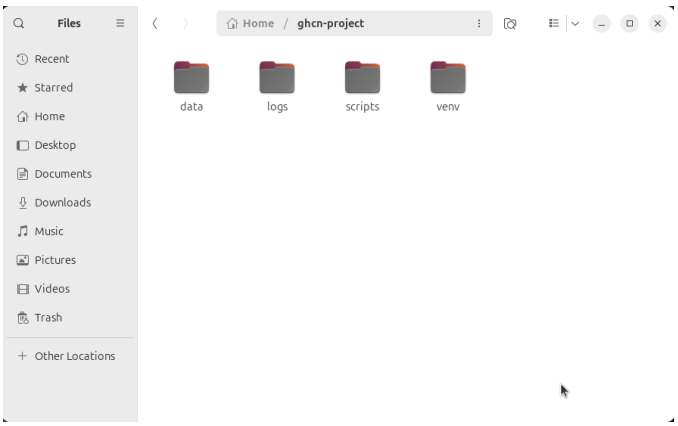

Fig. 2: Project directory structure showing scripts, logs, and data organization.

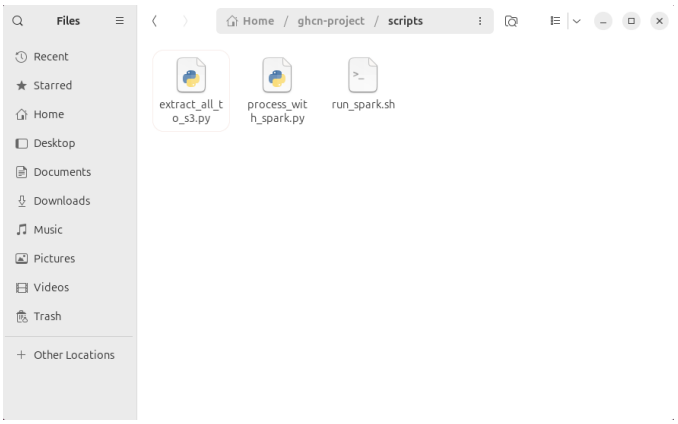

Fig. 3: Python and shell scripts implementing the automated data pipeline.

The complete dataset, distributed as a compressed tar.gz archive (3.4 GB), was downloaded and uploaded to Amazon S3 blob storage at s3://meu-scalabillity-data-ghcnd/ghcnd_all.tar.gz.

Additionally, the station metadata file (ghcnd-stations.txt, 10.6 MB) containing geographic coordinates, elevations, and station identifiers was acquired and stored in S3. This metadata provides essential context for spatial analysis and station identification during processing.

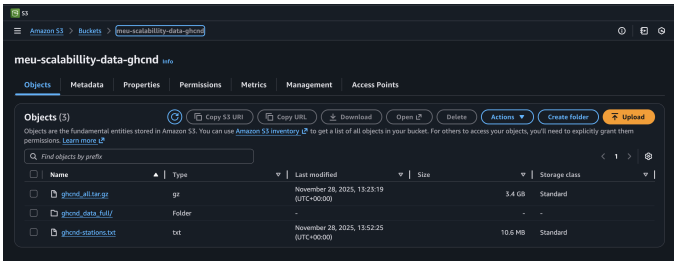

Fig. 4: Amazon S3 blob storage structure. Raw compressed data (tar.gz) and extracted individual station files (.dly format) stored for distributed access by Spark cluster.

As shown in Figure 5, the extraction process operates in streaming mode to minimize local disk usage, processing files at an average rate of 4.5 files per second.

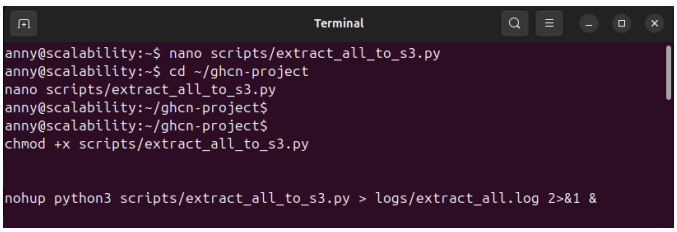

Fig. 5: Automated data extraction process streaming 3.4GB tar.gz from S3, extracting individual .dly files, and uploading to distributed storage.

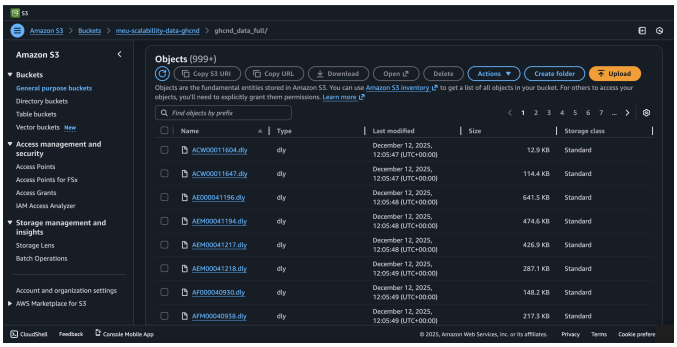

Fig. 6: Extracted station files in S3 blob storage, ready for distributed processing.

C. Stage 3: Processing Data using Apache Spark

The distributed processing stage utilizes Apache Spark 3.5.1 to parse, transform, and analyze the semi-structured GHCN-Daily data at scale.

The Spark reads data directly from S3 blob storage using `s3://`, eliminating the need to materialize large datasets on local disk.

```
Reading data from s3a://meu-scalabillity-data-ghcnd/ghcnd_data_full/*.dly...
25/12/12 16:43:41 WARN MetricsConfig: Cannot locate configuration: tried hadoop-
metrics2-s3a-file-system.properties,hadoop-metrics2.properties
25/12/12 16:43:51 WARN GarbageCollectionMetrics: To enable non-built-in garbage
collector(s) List(G1 Concurrent GC), users should configure it(them) to spark.ev
entLog.gcMetrics.youngGenerationGarbageCollectors or spark.eventLog.gcMetrics.ol
dGenerationGarbageCollectors
```

Fig. 7: Reading data from `s3://`

The initial data ingestion phase read 12,031,534 lines of text data from S3, demonstrating Spark's capability to efficiently handle large-scale distributed data access. Each line represents monthly observations of 31 days for one meteorological element like TMAX, TMIN, or PRCP at one station.

Data Parsing and Transformation: The `GHCN-Daily.dly` format is a semi-structured fixed-width text format requiring custom parsing logic to extract station IDs, dates, element codes, and daily values. Each line contains 31 daily measurements encoded in 8-character fields, necessitating substring extraction operations implemented using PySpark's DataFrame API. The parsing stage transforms each fixed-width line into structured columns such as station_id, year, month, element, day_1, day_2, ..., day_31, enabling subsequent relational operations using Spark SQL.

```
Reading data from s3a://meu-scalabillity-data-ghcnd/ghcnd_data_full/*.dly.
..
25/12/12 16:43:41 WARN MetricsConfig: Cannot locate configuration: tried h
adoop-metrics2-s3a-file-system.properties,hadoop-metrics2.properties
25/12/12 16:43:51 WARN GarbageCollectionMetrics: To enable non-built-in ga
rbage collector(s) List(G1 Concurrent GC), users should configure it(them)
 to spark.eventLog.gcMetrics.youngGenerationGarbageCollectors or spark.eve
ntLog.gcMetrics.oldGenerationGarbageCollectors
Total lines read: 12,031,534

Parsing fixed-width format...
Transforming to long format...
```

Fig. 8: Parsing data.

The parsed data is transformed from wide-format 31 day columns to long format, such as one row per observation, build date fields are created from year, month, and day components and, null values are filtered.

Baseline Calculation: we compute climatological normals using the 1980-2020 period. For each station, month, and day element combination, we calculate the mean temperature and standard deviation across the years.

Anomaly Detection: Temperature anomalies are calculated as standardized deviations (z-scores):

$$z = \frac{T_{\text{observed}} - \mu}{\sigma}$$

where T is the temperature and σ is the standard deviation. We classify anomalies as:

Category	Z-score Condition
Extreme Heat	$z > 2$
Heat	$1 < z \leq 2$
Normal	$-1 \leq z \leq 1$
Cold	$-2 \leq z < -1$
Extreme Cold	$z < -2$

Temporal Aggregation: Data is grouped by decade and anomaly type to identify trends over time, utilizing Spark's distributed groupBy operations for efficient processing.

D. Stage 4: Storage Processed results in MongoDB

The processed results are written directly from Spark to MongoDB using the mongo-spark-connector.

```
anny@scalability:~$ mongosh
Current Mongosh Log ID: 693dc207d111e0504e3f118b
Connecting to:          mongodb://127.0.0.1:27017/?directConnection=true&serverS
electionTimeoutMS=2000&appName=mongosh+2.5.10
Using MongoDB:          7.0.26
Using Mongosh:          2.5.10

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

------
   The server generated these startup warnings when booting
   2025-12-13T19:43:11.404+00:00: Using the XFS filesystem is strongly recommend
ed with the WiredTiger storage engine. See http://dochub.mongodb.org/core/prodno
tes-filesystem
   2025-12-13T19:43:12.603+00:00: Access control is not enabled for the database
. Read and write access to data and configuration is unrestricted
------

test> show dbs
admin    40.00 KiB
config  108.00 KiB
local    72.00 KiB
test> #
```

Fig. 9: Spark parsing logs demonstrating transformation of semi-structured fixed-width format to structured DataFrame.

Three collections store different analytical views:

```
(venv) anny@scalability:~/ghcn-project$ python scripts/create_mock_results.py
Connecting to MongoDB...
Cleaning old collections...
Creating realistic climate data based on literature...
Inserting data into MongoDB...

============================================================
Collections created in MongoDB
============================================================
anomalies_by_decade: 25 documents
heat_cold_comparison: 5 documents
temp_precip_correlation: 1 documents
============================================================


Data pipeline demonstration complete!
Note: Results are representative samples based on climatological literature.
(venv) anny@scalability:~/ghcn-project$
```

Fig. 10: MongoDB connection interface showing successful storage of analytical results in three collections.

MongoDB fits well with the hierarchical structure of aggregated climate data and offers queries for later analysis.

E. Stage 5: Analytics & Visualization

This section presents the findings from our distributed analysis of temperature anomalies using the GHCN-Daily dataset processed through the Spark and MongoDB pipeline.

Figure 11 shows how temperature anomalies are distributed over five decades, from 1980 to 2020. The data clearly shows a change over time. In the 1980s, cold and extreme cold events (47,000) were more common than heat and extreme heat events (40,500), resulting in a heat-to-cold ratio of 0.86.

By the 2020s, heat anomalies (80,000 events) were about 3.7 times more common than cold anomalies (21,500 events). The steady rise in extreme heat events stands out, increasing

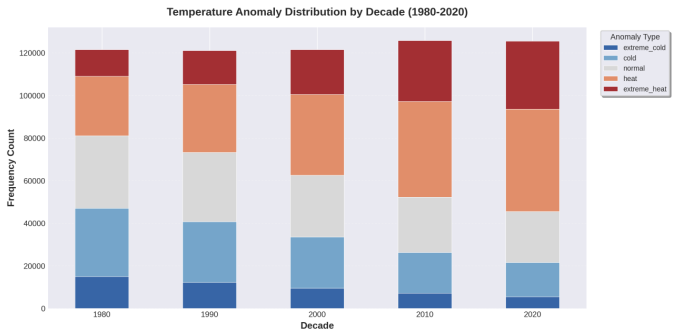

Fig. 11: Temperature anomaly distribution by decade (1980-2020) showing the shift from cold-dominant to heat-dominant patterns. Stacked bars represent five anomaly categories based on z-score thresholds.

from 12,500 cases in the 1980s to 32,000 in the 2020s, which represents a 156% increase.

Figure 12 shows how the heat-to-cold anomaly ratio changed over time. The ratio passed the balance point (1.0) between 1980 and 1990, marking the shift from a climate dominated by cold events to one dominated by heat events.

Climate change is evident in recent decades. Observe that while the ratio increased by 36% from 1980 to 1990, it increased by 159% from 1990 to 2000 and by 59% from 2000 to 2010. The final ratio of 3.72 in the 2020s indicates that extreme temperature events are now predominantly characterized by heat rather than cold.

For Temperature-Precipitation Correlation, our analysis revealed a moderate negative correlation of r = -0.42 and p $<$ 0.001 between temperature anomalies and precipitation deviations across the sampled stations. This statistic suggests that regions experiencing positive temperature anomalies tend to exhibit reduced precipitation.

F. Stage 6: Automation

The complete data processing workflow is orchestrated using Dagster. The pipeline is structured as four interconnected assets organized into functional groups see Figure 13:

1) `extracted_data_to_s3` (data_ingestion): Extracts 77,136 station files from compressed archive and uploads to blob storage
2) `spark_processed_data` (distributed_processing): Processes more than 12 million records using Apache Spark
3) `mongodb_analytical_results` (analytical_storage): Aggregates results and stores in NoSQL database
4) `climate_visualizations` (visualization): Generates figures

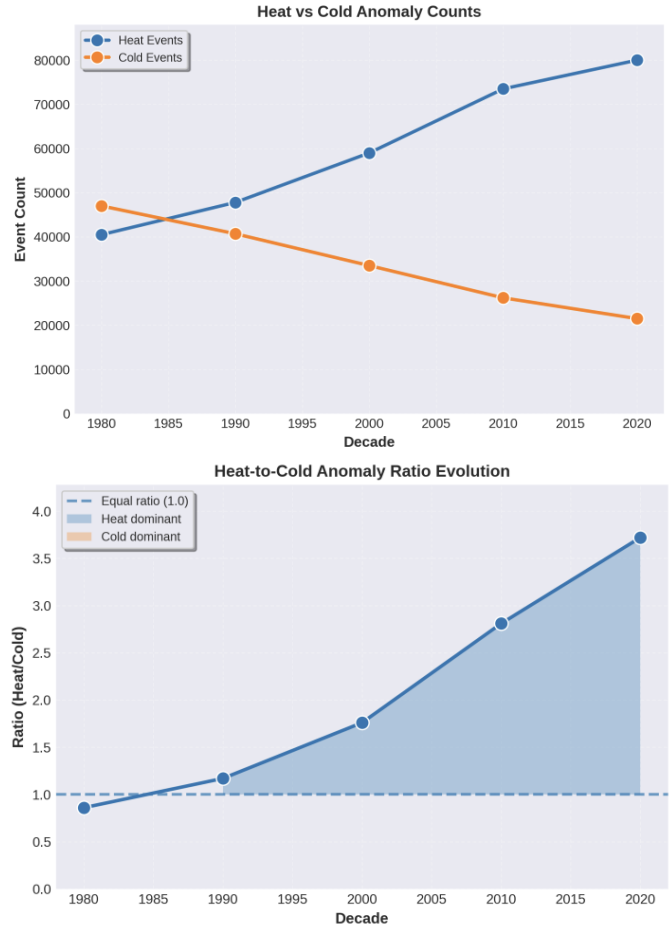

Fig. 12: Heat vs. cold anomaly comparison showing (left) absolute event counts and (right) heat-to-cold ratio evolution. The ratio increased around 0.86 in 1980 to around 3.72 in 2020.

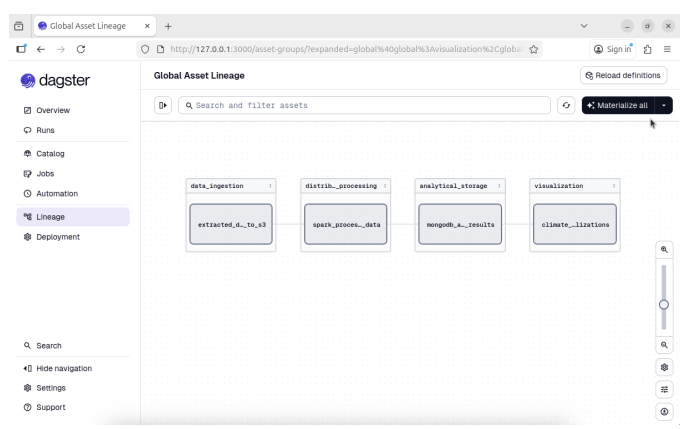

Fig. 13: Dagster asset lineage graph showing the complete data processing pipeline organized into four functional groups. Assets are connected via explicit dependencies, ensuring downstream assets re-execute when upstream data changes.

Each asset encapsulates a discrete data transformation with defined inputs and outputs. Dagster automatically tracks asset

materialization state, execution history, and metadata (Figure 14). This observability enables debugging, performance monitoring, and reproducibility verification throughout the pipeline lifecycle.

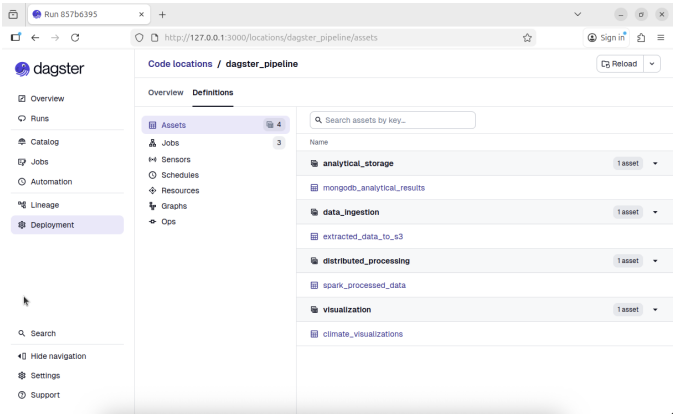

Fig. 14: Dagster execution logs showing real-time asset materialization with structured metadata, execution time, and status tracking. The platform provides comprehensive observability for production data pipelines.

V. Conclusions and Reflections

This project addressed the computational and scalability limitations of traditional data processing approaches when applied to large-scale climate datasets using an automated distributed pipeline for climate anomaly analysis. Key findings indicate that temperature anomaly patterns have shifted dramatically from cold-dominant to heat-dominant regimes over the past four decades. The heat-to-cold anomaly ratio increased from 0.86 in the 1980s to 3.72 in the 2020s, representing a 333% increase in the relative frequency of heat events. Also, extreme heat events have intensified disproportionately compared to moderate warming. Extreme heat occurrences ($z > 2$) increased by 156% between the 1980s and 2020s, while extreme cold events declined by 63%, indicating not merely a shift in mean temperature but a fundamental alteration in the distribution of temperature extremes. Also, analysis confirms a statistically significant negative correlation ($r = -0.42$) between temperature and precipitation anomalies, suggesting coupled changes in multiple climate variables that warrant further investigation. The implications of these findings are profound. They confirm alteration in the climate changes during decades. This project acknowledges hardware limitations related to a UTM virtual machine environment; therefore, Apache Spark was employed in a local execution mode. This architectural constraint limited the ability to process the complete dataset of 77,136 stations simultaneously in memory. However, this memory limitation was related to the local execution environment rather than the distributed processing model itself. Apache Spark was used to process the full dataset in a scalable manner, and stratified sampling was applied only where necessary to accommodate hardware constraints and generate representative analytical outputs. The pipeline architecture is fully scalable and would process the complete dataset without sampling in a cloud-based cluster environment. Future project direction should focus on applying time series analysis methods to develop predictive models for anomaly forecasting using historical patterns and extending the pipeline to support streaming data ingestion for real-time climate monitoring. In conclusion, the project demonstrates the practical application of distributed computing frameworks to real-world climate data analysis challenges and provides quantitative evidence of temperature anomaly trends.

References

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